

RTC-based Event Logger

ROBIN CHALANA

Event log data is a unique source of information for security, audit, compliance and troubleshooting purposes. An event logger enables you to view the log of a series of events. This can be done not only at the time of occurrence of the events but also any time later. The event logger provides an accurate time and date at which a particular event has occurred, which is essential for many applications.

In this event logger article, we also demonstrate the use of a DS1307 real-time clock (RTC) chip, switches and an LCD module. The RTC chip provides time components of hours, minutes and seconds in addition to the year, month and day.

In this project, pressing a switch is regarded as an event. When the

event occurs, it is recorded in memory locations of the microcontroller. This information from the microcontroller can be retrieved and seen on the LCD module and HyperTerminal.

Circuit and working

Fig. 1 shows the circuit of RTC-based event logger. It comprises regulator 7805 (IC1), microcontroller P89V51RD2 (IC2), RS232 interface MAX232 (IC4), RTC chip DS1307 (IC3), LCD module and a few discrete components.

Microcontroller. The P89V51RD2 is an 8-bit 80C51 microcontroller with 64kB Flash and 1024 bytes of data RAM. It has three 16-bit timers/counters, programmable watchdog timer, eight interrupt sources with four priority levels, serial peripheral interface (SPI) and enhanced UART, programmable counter array with PWM and capture/compare functions, and four 8-bit input/output (I/O) ports. It also has an on-chip oscillator and clock circuitry, which is operated at up to 40MHz crystal. It supports 12 clocks per machine cycle (default) and 6 clocks per machine cycle modes, which can be selected via software. The Flash program memory supports both parallel programming and serial programming.



Real-time clock chip. DS1307 is a serial RTC chip with calendar function. This chip also incorporates 56 bytes of non-volatile RAM. Data and address are transferred serially through I²C bi-directional bus, which obviates the need for a large number of interface lines. In this chip, the clock operates in either 24-hour or 12-hour format with AM/PM indicator. In calendar mode, end of the month is automatically adjusted for months with less than 31 days and leap year compensation. The memory map of DS1307 is shown in Table I.

Clock pulse to the RTC is provided by a 32.678kHz crystal. As per convention of I²C interface communication, device address while writing into the RTC chip is 0xD0 and device address while reading data from the RTC chip is 0xD1.

Using SCL and SDA lines, the microcontroller can read/write data from/to the memory of the RTC chip.

Some of the conditions to communicate through the I²C bus are:

Clock and data transition. The SDA line is normally pulled high with an external device. Data on the SDA line can change only when SCL line is low. Data changes during a high SCL line

PARTS LIST

Semiconductors:

IC1	- 7805, 5V regulator
IC2	- P89V51RD2 microcontroller
IC3	- DS1307 RTC
IC4	- MAX232 RS232 driver
D1-D6	- 1N4007 rectifier diode
LCD	- 16×2-line LCD module
LED1	- 5mm LED

Resistors (all ¼-watt, ±5 per cent carbon, unless stated otherwise):

R1	- 330-ohm
R2-R7	- 10-kilo-ohm
R8	- 120-ohm, 1W
VR1	- 10-kilo-ohm preset
RN1	- 10-kilo-ohm resistor network

Capacitors:

C1	- 1000µF, 35V electrolytic
C2	- 0.1µF ceramic disk
C3	- 10µF, 16V electrolytic
C6-C10	- 1µF, 16V electrolytic
C4, C5	- 33pF ceramic disk

Miscellaneous:

X1	- 230V AC primary to 12V, 500mA secondary transformer
S1-S20	- Push-to-on tactile switch
X _{crystal}	- 11.0592MHz crystal
X _{rtc}	- 32.678kHz crystal
BATT1	- 3.3V battery
BATT2	- 9V battery
CON1	- Serial COM port connector

TABLE I
Memory Map of DS1307

Address	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Function	Range
00H	CH	10 seconds			Seconds				Seconds	00-59
01H	0	10 minutes			Minutes				Minutes	00-59
02H	0	12	10 hours	10 hours	Hours				Hours	01-12
		24	PM/AM							00-23
03H	0	0	0	0	0	Day				01-07
04H	0	0	10 date			Date			Date	01-31
05H	0	0	0	10 months	Month				Month	01-12
06H	10 years			Year					Year	00-99
07H	Out	0	0	SQWE	0	0	RS1	RS0	Control	—
08H-3FH									RAM 56x8	00-FFH